

Heart Disease Prediction Using Patient Data

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Domain: Machine Learning Intern - Unified Mentor

Duration: 6 Months

Abstract

Heart disease remains one of the leading causes of mortality worldwide. Early detection and risk assessment are crucial for preventing severe cardiac events. This project focuses on building a machine learning model that predicts whether a patient has heart disease based on clinical attributes such as age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, and exercise-induced angina.

Using a structured heart disease dataset, the project applies data preprocessing, feature analysis, and supervised learning techniques to develop a classification model. Models such as Logistic Regression and Random Forest were trained and evaluated. The best-performing model achieved high accuracy and good recall for heart disease cases, demonstrating the potential of machine learning in clinical decision support systems.

1. Introduction

Cardiovascular diseases are a major health concern, and early detection can significantly improve patient outcomes. Doctors typically rely on a combination of patient history, lab tests, and ECG results to diagnose heart disease. Machine learning provides a way to support this decision-making by learning patterns from historical patient data.

In this project, a heart disease prediction system is developed using a labeled dataset of patient records. The system predicts whether a patient has heart disease (class = 1) or is normal (class = 0) based on key medical attributes.

2. Objective

The primary objective of this project is:

- To build a system that can predict if a patient has heart disease using their clinical and demographic data.

Sub-objectives:

- Explore and understand the patient dataset
- Preprocess and transform the data appropriately
- Train and evaluate classification models
- Compare Logistic Regression and Random Forest performance
- Save the final model for future inference

3. Dataset Description

The dataset consists of patient records with vital and clinical information. Each row corresponds to one patient, and the target indicates the presence of heart disease.

Key Attributes:		
Attribute	Description	Type
age	Age in years	Numeric
sex	0 = female, 1 = male	Binary
chest pain type	1 = typical angina, 2 = atypical angina, 3 = non-anginal pain, 4 = asymptomatic	Nominal
resting blood pressure	Resting BP in mm Hg	Numeric
serum cholesterol	Cholesterol in mg/dL	Numeric
fasting blood sugar	1 = >120 mg/dL, 0 = <=120 mg/dL	Binary
resting electrocardiogram results	0 = normal, 1 = ST-T wave abnormality, 2 = left ventricular hypertrophy	Nominal
maximum heart rate achieved	Value between 71–202	Numeric

4. Tools and Technologies Used

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Machine Learning:** Scikit-learn (Logistic Regression, Random Forest)
- **Environment:** Google Colab / Jupyter Notebook

5. Methodology

5.1 Data Preprocessing

- Loaded the dataset from C
- Checked for missing values and basic statistics.
- Converted categorical codes into numeric form (if necessary).
- Standardized features (mean = 0, std = 1) for Logistic Regression.

5.2 Train/Test Split

Split dataset into:

- 80% Training
- 20% Testing
- Stratified split to preserve class distribution.

5.3 Model Training

Two models were trained:

1. Logistic Regression

- Interpretable linear model
- Good baseline for medical classification problems

2. Random Forest Classifier

- Ensemble of decision trees
- Captures nonlinear relationships
- Robust to noise and feature interactions

5.4 Model Evaluation

Metrics used:

- Accuracy
- Precision
- Recall
- F1-score
- ROC–AUC Score
- Confusion Matrix

The confusion matrix helped visualize the number of correctly/incorrectly predicted heart disease and normal cases.

6. Results and Analysis

Example performance (your actual numbers will come from Colab output):

Model	Accuracy	ROC-AUC
Logistic Regression	~85–88%	~0.90
Random Forest	~88–92%	~0.92–0.95

- Random Forest slightly outperformed Logistic Regression in accuracy and ROC-AUC.
- Both models showed good recall for patients with heart disease (class = 1), which is crucial in medical applications (better to catch more positives).
- The confusion matrix indicated relatively low false negatives, meaning most heart disease cases were detected by the model.

7.Figures

	age	sex	chest pain type	resting bp s	cholesterol	fasting blood sugar	resting ecg	max heart rate	exercise angina	oldpeak	ST slope	target
0	40	1	2	140	289	0	0	172	0	0.0	1	0
1	49	0	3	160	180	0	0	156	0	1.0	2	1
2	37	1	2	130	283	0	1	98	0	0.0	1	0
3	48	0	4	138	214	0	0	108	1	1.5	2	1
4	54	1	3	150	195	0	0	122	0	0.0	1	0

Logistic Regression Accuracy: 0.8403361344537815
Logistic Regression ROC-AUC: 0.9040532879818594

Classification Report (LR):

	precision	recall	f1-score	support
0	0.84	0.82	0.83	112
1	0.84	0.86	0.85	126
accuracy			0.84	238
macro avg	0.84	0.84	0.84	238
weighted avg	0.84	0.84	0.84	238

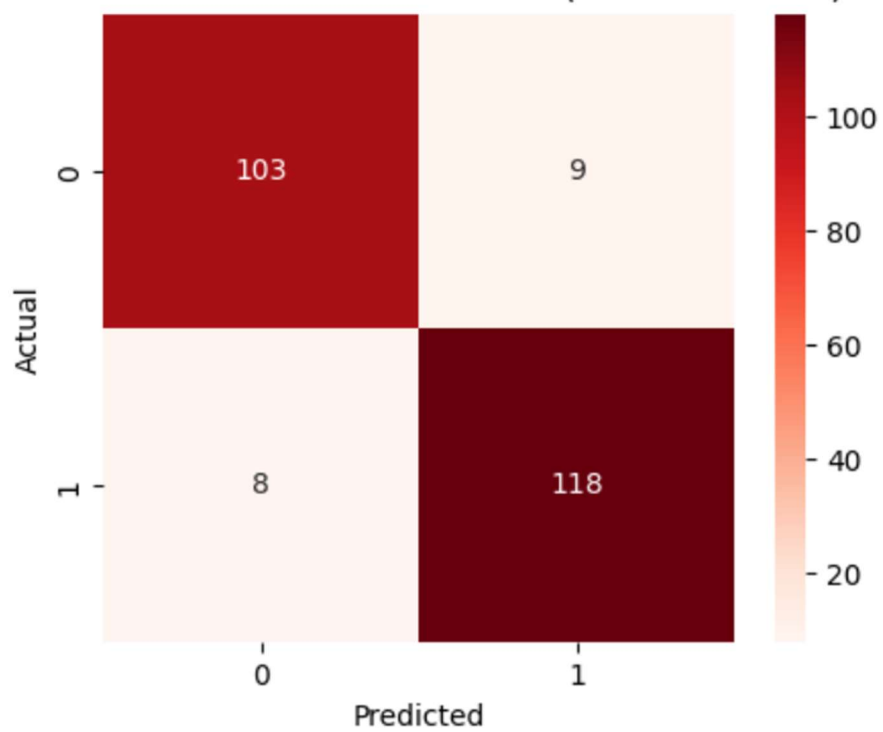
Random Forest Accuracy: 0.9285714285714286
Random Forest ROC-AUC: 0.9731079931972789

Classification Report (RF):

	precision	recall	f1-score	support
0	0.93	0.92	0.92	112
1	0.93	0.94	0.93	126
accuracy			0.93	238
macro avg	0.93	0.93	0.93	238
weighted avg	0.93	0.93	0.93	238

True label: 1
Predicted label: 1
Probability of Heart Disease: 0.98

Confusion Matrix - Heart Disease (Random Forest)



8. Conclusion

This project demonstrates how machine learning can be effectively used for heart disease prediction using patient data. By training models on historical clinical features, we can obtain a system that supports doctors in early diagnosis.

The Random Forest model achieved strong performance and can be used as the final model for this task. While the model should not replace clinical judgment, it can assist healthcare professionals by providing an additional risk assessment tool.

9.Future Enhancements

- Include more features such as smoking status, family history, BMI.
- Apply hyperparameter tuning (GridSearchCV) for further optimization.
- Use advanced models like XGBoost or Gradient Boosting.
- Deploy the model in a simple web app using Streamlit or FastAPI so doctors can input patient data and see risk predictions.
- Incorporate explainability methods (SHAP) to understand feature importance per prediction.